

Eksamen
Kopiereg voorbehou

Examination
Copyright reserved

Module EBN111
Elektrisiteit en Elektronika
8 June 2009

Module EBN111
Electricity and Electronics
8 June 2009



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute 180 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	26		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Vrae van Afdelings A en B moet op die ANTWOORDBLAD beantwoord word.
Questions from Sections A and B must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the exam paper – the exam paper will not be taken in.
- Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$
Polar form: $A\angle\theta$; Rectangular form: $a + jb$

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 D Bhatt
Groep hoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>	
Groep hoof: <i>Group Head:</i>	Prof J Joubert
	Handtekening: <i>Signature:</i>

Afdeling A

Section A

Vrae van hierdie afdeling moet op die ANTWOORDBLAD beantwoord word. Kies die mees korrekte antwoord.

Questions from this section should be answered on the ANSWER SHEET. Choose the answer which is most correct.

VRAAG/QUESTION 1 [30]

Vraag 1.1 [1]

Sien Fig. Q1.1. In die stroombaan getoon sal 'n verlaging in R_3 lei tot 'n verhoging van:

- A) Die stroom deur R_2 .
- B) Die spanning oor R_1 .
- C) Die spanning oor R_3 .
- D) V_s .
- E) Geen van bogenoemde nie.

Question 1.1 [1]

See Fig. Q1.1. In the circuit shown, a decrease in R_3 leads to a increase of:

- A) The current through R_2 .
- B) The voltage across R_1 .
- C) The voltage across R_3 .
- D) V_s .
- E) None of the above.

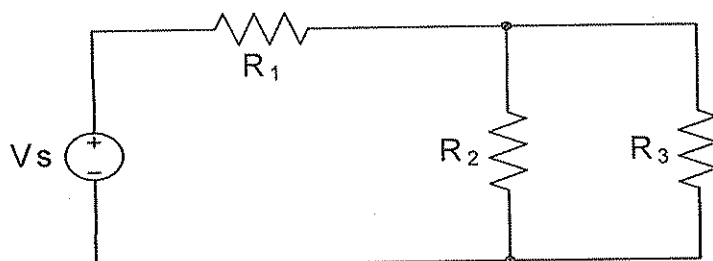


Fig. Q1.1

Vraag 1.2 [1]

Sien Fig. Q1.2. Vir die stroombaan getoon, wat is die waarde van G_{eq} ?

- A) 14.2 S
- B) 4.36 S
- C) 3.27 S
- D) 22 S
- E) Geen van bogenoemde nie.

Question 1.2 [1]

See Fig. Q1.2. For the circuit shown, what is the value of G_{eq} ?

- A) 14.2 S
- B) 4.36 S
- C) 3.27 S
- D) 22 S
- E) None of the above.

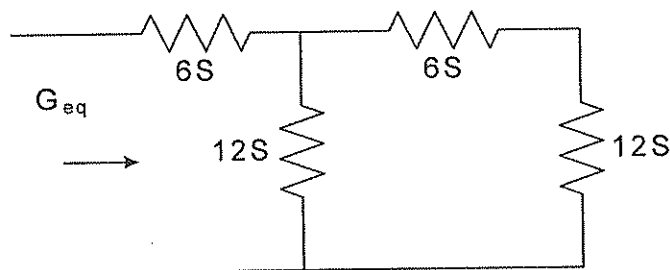


Fig. Q1.2

Vraag 1.3 [1]

As drywing vir 'n element gemeet word mbv 'n voltmeter en 'n ammeter, moet:

- A) Die ammeter in serie en die voltmeter in parallel verbind word.
- B) Die voltmeter in serie en die ammeter in parallel verbind word.
- C) Beide in parallel verbind word.
- D) Beide in serie verbind word.
- E) Geen van bogenoemde nie.

Question 1.3 [1]

When measuring power in an element by using a voltmeter and an ammeter:

- A) The ammeter must be connected in series and the voltmeter in parallel.
- B) The voltmeter must be connected in series and the ammeter in parallel.
- C) Both should be connected in parallel.
- D) Both should be connected in series.
- E) None of the above.

Vraag 1.4 [1]

Sien Fig. Q1.4. Watter van die volgende bevredig nie Kirchhoff se Spanningswet nie?

Question 1.4 [1]

See Fig. Q1.4. Which of the following circuits does not satisfy Kirchhoff's Voltage Law?

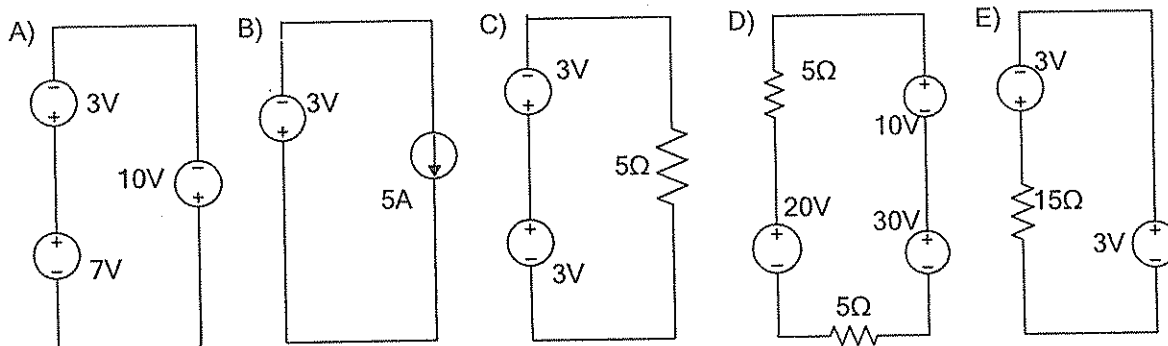


Fig. Q1.4

Vraag 1.5 [1]

Sien Fig. Q1.5. Vir die stroombaan getoon:

- A) $I_1 = -10\text{A}$; $I_2 = 10\text{A}$
- B) $I_1 = 12\text{A}$; $I_2 = -10\text{A}$
- C) $I_1 = 12\text{A}$; $I_2 = 7\text{A}$
- D) $I_1 = -10\text{A}$; $I_2 = 13\text{A}$
- E) Geen van bogenoemde nie

Question 1.5 [1]

See Fig. Q1.5. For the circuit shown:

- A) $I_1 = -10\text{A}$; $I_2 = 12\text{A}$
- B) $I_1 = 12\text{A}$; $I_2 = -10\text{A}$
- C) $I_1 = 12\text{A}$; $I_2 = 7\text{A}$
- D) $I_1 = -10\text{A}$; $I_2 = 13\text{A}$
- E) None of the above

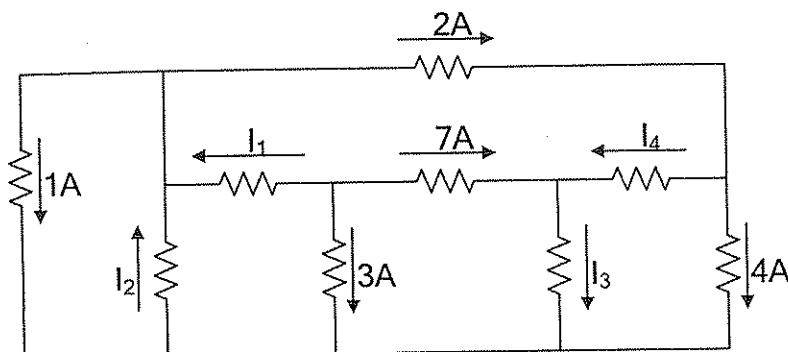


Fig. Q1.5

Vraag 1.6 [1]

Watter van die volgende stellings oor Kirchoff se stroomwet is vals.

- A) Algebraïse som van strome by 'n node is nul.
- B) Algebraïse som van die strome by 'n geslote grens is nul.
- C) Som van strome wat ingaan by 'n node is gelyk aan die som van strome wat uitgaan.
- D) Som van strome wat ingaan by 'n node is gelyk aan die negatief van die som van strome wat uitgaan.
- E) Geen van bogenoemde nie.

Question 1.6 [1]

Which of the following statement about Kirchoffs Current law is false.

- A) Algebraic sum of currents at a node is zero.
- B) Algebraic sum of currents at a closed boundary is zero.
- C) Sum of currents entering a node is equal to sum of currents leaving a node.
- D) Sum of currents entering a node is equal to the negative of the sum of currents exiting node.
- E) None of the above.

Vraag 1.7 [1]

Die stroom deur 'n element word getoon in Fig. Q1.7. Bepaal die totale lading wat deur die element gevloei het by $t = 6$. Aanvaar dat geen stroom gevloei het vir $t < 0$ nie.

- A) -20 C
- B) -75 C
- C) 20 C
- D) 45 C
- E) 75 C

Question 1.7 [1]

The current through an element is shown in Fig. Q1.7. Determine the total charge that passed through the element at $t = 6$. Assume that no current passed through the element for $t < 0$.

- A) -20 C
- B) -75 C
- C) 20 C
- D) 45 C
- E) 75 C

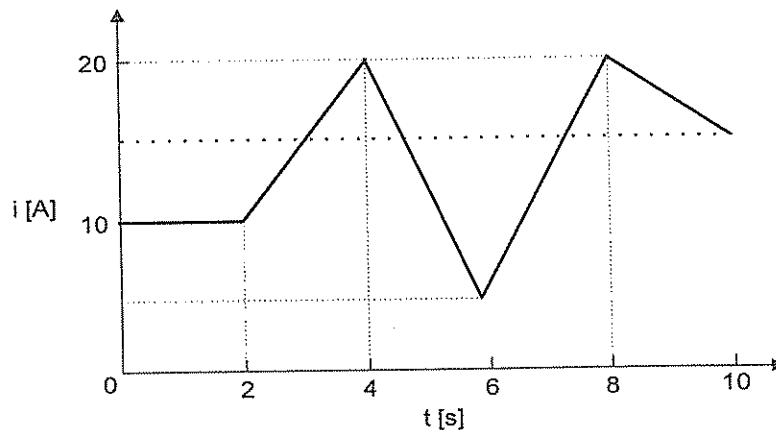


Fig. Q1.7

Vraag 1.8 [1]

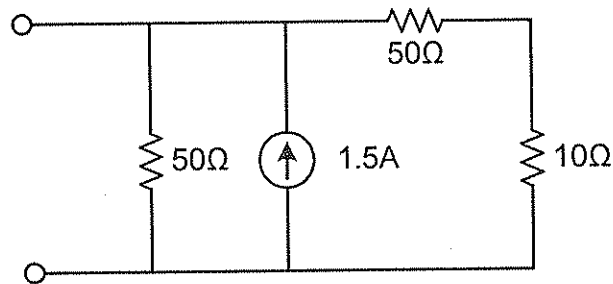
Vraag 1.8 verwys na Fig. Q1.8. Bepaal die waarde van 'n resistor wat oor die oop terminale geplaas kan word wat maksimum drywingsoordrag sal verseker.

- A) $110\ \Omega$
- B) $27.27\ \Omega$
- C) $50\ \Omega$
- D) $8.11\ \Omega$
- E) Geen van bogenoemde nie

Question 1.8 [1]

Question 1.8 refers to Fig. Q1.8 Determine the value of a resistor placed across the open terminals resulting in maximum power transferred to the resistor.

- A) $110\ \Omega$
- B) $27.27\ \Omega$
- C) $50\ \Omega$
- D) $8.11\ \Omega$
- E) None of the above

**Fig. Q1.8****Vraag 1.9 [1]**

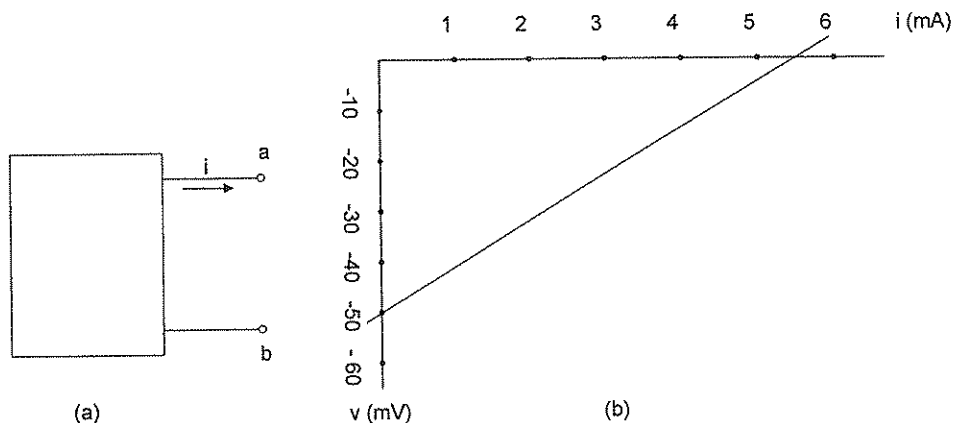
Die spannings-stroom karakteristieke in Fig Q1.9b is verkry vanaf die onbekende lineêre stroombaan in Fig Q1.9a. Die baan bestaan slegs uit afhanklike en onafhanklike spanningsbronne, stroombronne en weerstande. Bepaal die opebaan spanning V_{OC} , en die kortsluit-stroom I_{SC} ?

- A) $V_{OC} = 0\text{ V}$, $I_{SC} = 0\text{ A}$
- B) $V_{OC} = 50\text{ V}$, $I_{SC} = 5.5\text{ A}$
- C) $V_{OC} = -50\text{ V}$, $I_{SC} = -5.5\text{ A}$
- D) $V_{OC} = -50\text{ mV}$, $I_{SC} = 5.5\text{ mA}$
- E) $V_{OC} = -50\text{ mV}$, $I_{SC} = 55\text{ mA}$
- F) Geen van bogenoemde nie

Question 1.9 [1]

The voltage-current characteristics in Fig Q1.9b were obtained from the unknown linear circuit in Fig Q1.9a. The circuit consists only of dependent and independent voltage sources, current sources and resistors. What is the open circuit voltage V_{OC} , and the short circuit current I_{SC} ?

- A) $V_{OC} = 0\text{ V}$, $I_{SC} = 0\text{ A}$
- B) $V_{OC} = 50\text{ V}$, $I_{SC} = 5.5\text{ A}$
- C) $V_{OC} = -50\text{ V}$, $I_{SC} = -5.5\text{ A}$
- D) $V_{OC} = -50\text{ mV}$, $I_{SC} = 5.5\text{ mA}$
- E) $V_{OC} = -50\text{ mV}$, $I_{SC} = 55\text{ mA}$
- F) None of the above

**Fig. Q1.9**

Vraag 1.10 [2]

Sien Fig. Q1.10 Bepaal V_o as $V_x = 10\text{V}$ en 25V .

- A) $V_o = 3.66\text{V}$ and 9.9V
- B) $V_o = -3.66\text{V}$ and -9.9V
- C) $V_o = -3.636\text{V}$ and 9.09V
- D) $V_o = 3.636\text{V}$ and 9.09V
- E) Geen van bogenoemde nie

Question 1.10 [2]

See Fig. Q1.10 Determine V_o when $V_x = 10\text{V}$ and 25V .

- A) $V_o = 3.66\text{V}$ and 9.9V
- B) $V_o = -3.66\text{V}$ and -9.9V
- C) $V_o = -3.636\text{V}$ and 9.09V
- D) $V_o = 3.636\text{V}$ and 9.09V
- E) None of the above

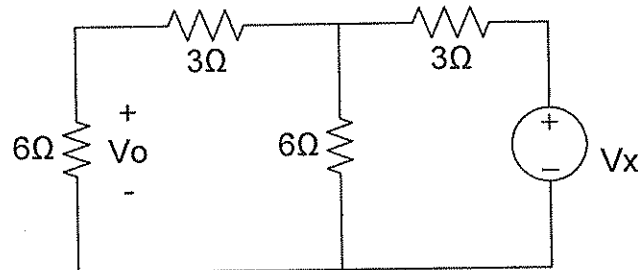


Fig. Q1.10

Vraag 1.11 [2]

Sien Fig. Q1.11. Bepaal die Thévenin ekwivalent?

- A) $R_{TH} = 12\ \Omega$, $V_{TH} = 6\text{V}$
- B) $R_{TH} = 5\ \Omega$, $V_{TH} = 6\text{V}$
- C) $R_{TH} = 6\ \Omega$, $V_{TH} = 3\text{V}$
- D) $R_{TH} = 3\ \Omega$, $V_{TH} = 6\text{V}$
- E) $R_{TH} = 3\ \Omega$, $V_{TH} = -6\text{V}$
- F) Geen van bogenoemde nie

Question 1.11 [2]

See Fig. Q1.11. What are the Thévenin equivalent?

- A) $R_{TH} = 12\ \Omega$, $V_{TH} = 6\text{V}$
- B) $R_{TH} = 5\ \Omega$, $V_{TH} = 6\text{V}$
- C) $R_{TH} = 6\ \Omega$, $V_{TH} = 3\text{V}$
- D) $R_{TH} = 3\ \Omega$, $V_{TH} = 6\text{V}$
- E) $R_{TH} = 3\ \Omega$, $V_{TH} = -6\text{V}$
- F) None of the above

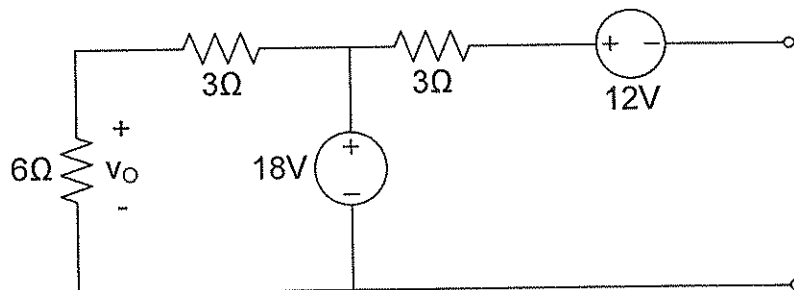


Fig. Q1.11

Vraag 1.12 [2]

Sien Fig. Q1.12. Die gloeilamp het 'n spesifikasie van 240V, 1.25A. Vind I_s sodat die gloeilamp werk by sy spesifikasiewaardes.

- A) 1.6 A
- B) -1.6 A
- C) 2.85 A
- D) -2.85 A
- E) Geen van bogenoemde nie

Question 1.12 [2]

See Fig. Q1.12. The light bulb is rated 240V, 1.25A. Find I_s to make the light bulb operate at its rated condition.

- A) 1.6 A
- B) -1.6 A
- C) 2.85 A
- D) -2.85 A
- E) None of the above

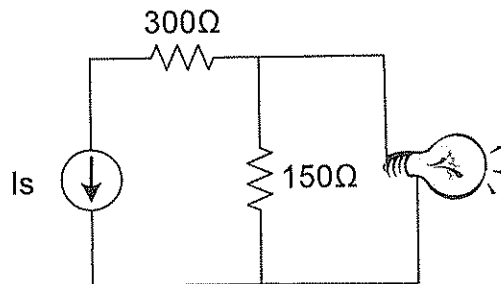


Fig. Q1.12

Vraag 1.13 [1]

Die ooplus-versterking van 'n nie-ideale operasionele versterker is:

- A) Oneindig
- B) Klein
- C) Baie groot
- D) 0
- E) Afhanklik van die inset spannings
- F) Geen van bogenoemde nie

Question 1.13 [1]

The open-loop gain of a non-ideal operational amplifier is:

- A) Infinite
- B) Small
- C) Extremely large
- D) 0
- E) Dependant on input voltages
- F) None of the above

Vraag 1.14 [1]

Die intree-weerstande van 'n ideale operasionele versterker is:

- A) Albei nul.
- B) Albei oneindig.
- C) Die omkeer inset is nul en die nie-omkeer inset is oneindig.
- D) Die omkeer inset is oneindig en die nie-omkeer inset is nul.
- E) Die omkeer inset is baie groot en die nie-omkeer inset is nul.
- F) Geen van bogenoemde nie

Question 1.14 [1]

The input resistances of an ideal operational amplifier are:

- A) Both zero.
- B) Both infinite.
- C) The inverting input is zero and the non-inverting input is infinite.
- D) The inverting input is infinite and the non-inverting input is zero.
- E) The inverting input is very large and the non-inverting input is zero.
- F) None of the above.

Vraag 1.15 [1]Sien Fig. Q1.15. Bepaal die waarde van v_o .

- A) 2 V
 B) -2 V
 C) 6 V
 D) -6 V
 E) 4 V
 F) -4 V
 G) Geen van bogenoemde nie

Question 1.15 [1]See Fig. Q1.15. Calculate the value of v_o .

- A) 2 V
 B) -2 V
 C) 6 V
 D) -6 V
 E) 4 V
 F) -4 V
 G) None of the above

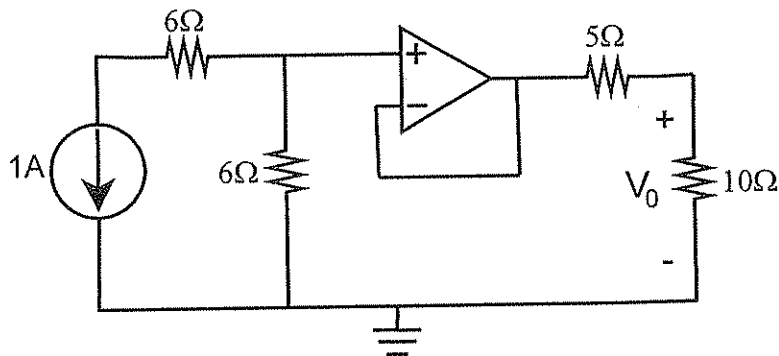


Fig. Q1.15

Vraag 1.16 [1]Sien Fig. Q1.16. Bepaal v_c .

- A) 24 V
 B) 12 V
 C) 0 V
 D) 16 V
 E) 8 V
 F) Geen van bogenoemde nie

Question 1.16 [1]See Fig. Q1.16. Calculate the value of v_c .

- A) 24 V
 B) 12 V
 C) 0 V
 D) 16 V
 E) 8 V
 F) None of the above

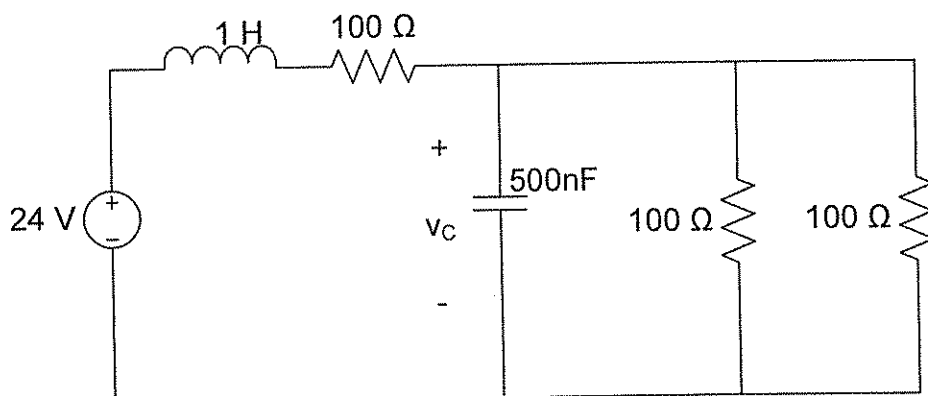


Fig. Q1.16

Vraag 1.17 [1] Watter een van die stellings oor fasors is vals? A) As 'n sinusodale funksie gedifferensieer word is dit dieselfde as om sy fasor met $j\omega$ te vermenigvuldig. B) As 'n sinusodale funksie geïntegreer word is dit dieselfde as om sy fasor met $j\omega$ te deel. C) As twee sinusodale funksies opgetel word is dit dieselfde as om hulle fasors op te tel. D) 'n Fasor is 'n komplekse getal wat die amplitude en frekwensie van 'n sinusodale funksie aandui. E) Geen van bogenoemde nie	Question 1.17 [1] Which one of the statements regarding phasors is false? A) Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$ B) Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$ C) Adding sinusoids of the same frequency is equivalent to adding their corresponding phasors. D) A phasor is a complex number that represents the amplitude and frequency of a sinusoid E) None of the above
--	---

Vraag 1.18 [1] Skryf die sinusoidale funksie as 'n fasor: $v(t) = 10\sin(2t + 120^\circ)$ V A) $V = 10 \angle 30^\circ$ B) $V = 10 \angle -30^\circ$ C) $V = 10 \angle 210^\circ$ D) $V = 10 \angle -210^\circ$ E) $V = 10 \angle 120^\circ$ F) Geen van bogenoemde nie	Question 1.18 [1] Express the sinusoid as a phasor: $v(t) = 10\sin(2t + 120^\circ)$ V A) $V = 10 \angle 30^\circ$ B) $V = 10 \angle -30^\circ$ C) $V = 10 \angle 210^\circ$ D) $V = 10 \angle -210^\circ$ E) $V = 10 \angle 120^\circ$ F) None of the above.
--	---

Vraag 1.19 [1] Evalueer die volgende komplekse getal en skryf die antwoord in polêre vorm: $\frac{5 \angle 60^\circ}{3 - j4}$ A) $1 \angle 113.13^\circ$ B) $1 \angle 6.87^\circ$ C) $1 \angle -66.87^\circ$ D) $1 \angle -113.13^\circ$ E) $1 \angle 66.87^\circ$ F) Geen van bogenoemde nie	Question 1.19 [1] Evaluate the following complex number and express in polar form: $\frac{5 \angle 60^\circ}{3 - j4}$ A) $1 \angle 113.13^\circ$ B) $1 \angle 6.87^\circ$ C) $1 \angle -66.87^\circ$ D) $1 \angle -113.13^\circ$ E) $1 \angle 66.87^\circ$ F) None of the above
--	--

Vraag 1.20 [1] Evalueer die volgende komplekse getal en skryf die antwoord in polêre vorm: $\frac{5 \angle 60^\circ}{-3 + j4}$ A) $1 \angle 113.13^\circ$ B) $1 \angle 6.87^\circ$ C) $1 \angle -66.87^\circ$ D) $1 \angle -113.13^\circ$ E) $1 \angle 66.87^\circ$ F) Geen van bogenoemde nie	Question 1.20 [1] Evaluate the following complex number and express in polar form: $\frac{5 \angle 60^\circ}{-3 + j4}$ A) $1 \angle 113.13^\circ$ B) $1 \angle 6.87^\circ$ C) $1 \angle -66.87^\circ$ D) $1 \angle -113.13^\circ$ E) $1 \angle 66.87^\circ$ F) None of the above
---	---

Vraag 1.21 [1]

Die impedansie en admittansie van 'n 1mF kapasitor, wat teen $\omega = 100$ rad/s werk, is:

- A) $0.1 \angle -90^\circ \Omega$ and $10 \angle 90^\circ S$
- B) $10 \angle -90^\circ \Omega$ and $0.1 \angle 90^\circ S$
- C) $0.1 \angle 90^\circ \Omega$ and $10 \angle -90^\circ S$
- D) $10 \angle 90^\circ \Omega$ and $0.1 \angle -90^\circ S$
- E) Geen van bogenoemde nie

Question 1.21 [1]

The impedance and admittance of a 1mF capacitor operating at $\omega = 100$ rad/s are:

- A) $0.1 \angle -90^\circ \Omega$ and $10 \angle 90^\circ S$
- B) $10 \angle -90^\circ \Omega$ and $0.1 \angle 90^\circ S$
- C) $0.1 \angle 90^\circ \Omega$ and $10 \angle -90^\circ S$
- D) $10 \angle 90^\circ \Omega$ and $0.1 \angle -90^\circ S$
- E) None of the above

Vraag 1.22 [2]

Sien Fig. Q1.22. Bepaal die fasorstroom I_0 .

- A) $4 \angle 0^\circ A$
- B) $2.4 \angle -90^\circ A$
- C) $0.6 \angle 0^\circ A$
- D) $-1 A$
- E) Geen van bogenoemde nie

Question 1.22 [2]

See Fig. Q1.22. Determine the phasor current I_0 .

- A) $4 \angle 0^\circ A$
- B) $2.4 \angle -90^\circ A$
- C) $0.6 \angle 0^\circ A$
- D) $-1 A$
- E) None of the above

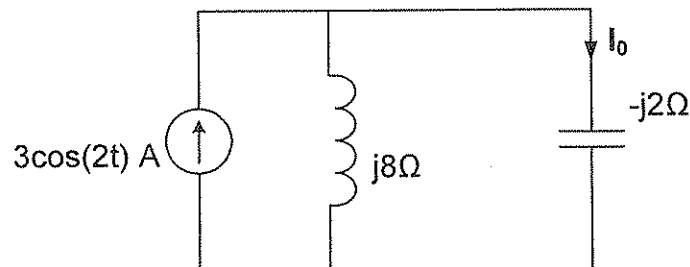


Fig. Q1.22

Vraag 1.23 [2]

Sien Fig. Q1.23. Bereken V_0 .

- A) $3.58 \angle 153.43^\circ A$
- B) $3.58 \angle 26.57^\circ A$
- C) $3.58 \angle -153.43^\circ A$
- D) $3.58 \angle -26.57^\circ A$
- E) Geen van bogenoemde nie

Question 1.23 [2]

See Fig. Q1.23. Determine V_0 .

- A) $3.58 \angle 153.43^\circ A$
- B) $3.58 \angle 26.57^\circ A$
- C) $3.58 \angle -153.43^\circ A$
- D) $3.58 \angle -26.57^\circ A$
- E) None of the above

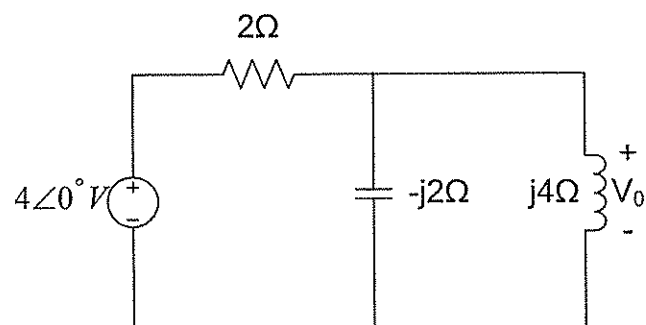


Fig. Q1.23

Vraag 1.24 [2]

Sien Fig. Q1.24. Die Norton ekwivalente impedansie by terminale a-b is.

- A) $10 + j6.67 \Omega$
- B) $10 - j6.67 \Omega$
- C) $10 - j20 \Omega$
- D) $10 + j20 \Omega$
- E) Geen van bogenoemde nie

Question 1.24 [2]

See Fig. Q1.24. The Norton equivalent impedance at terminals a-b is:

- A) $10 + j6.67 \Omega$
- B) $10 - j6.67 \Omega$
- C) $10 - j20 \Omega$
- D) $10 + j20 \Omega$
- E) None of the above

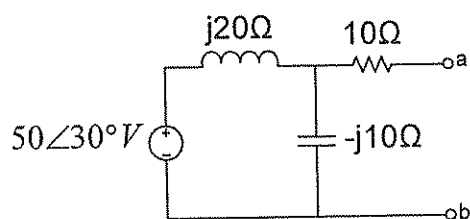


Fig. Q1.24

Afdeling B

Section B

Die finale antwoorde van vrae van hierdie afdeling moet na die ANTWOORDBLAD oorgedra word.

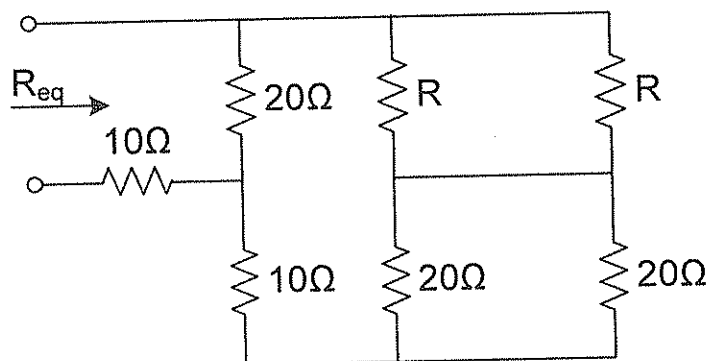
The final answers of questions from this section should be transferred to the ANSWER SHEET.

VRAAG 1 – QUESTION 1 [30]

Vraag 1.1	Question 1.1
a) Bepaal die strome i_1 en i_2. [2] b) Bepaal die spannings v_1 en v_2. [2] c) Bepaal die drywing (p) geabsorbeer (+) of voorsien (-) deur die 3A stroombron. [1]	a) Determine the currents i_1 and i_2. [2] b) Determine the voltages v_1 and v_2. [2] c) Find the power (p) absorbed (+) or supplied (-) by the 3A current source. [1]

Vraag 1.2

Question 1.2



Bepaal

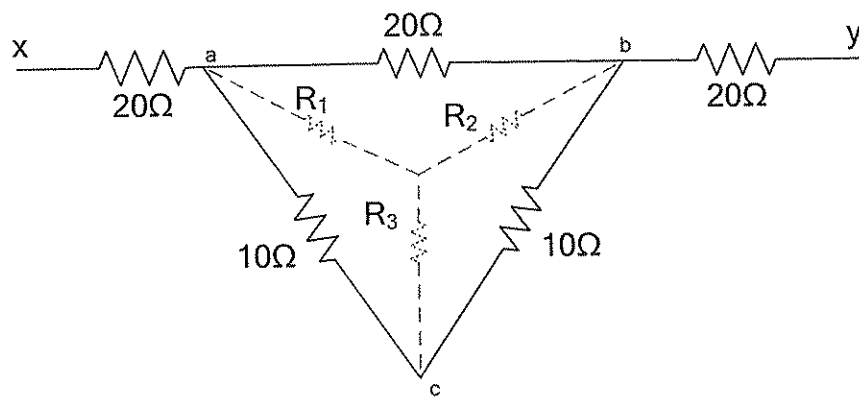
- a) R_{eq} , as $R = 0$ [1]
- b) R_{eq} as $R = \infty$ [1]
- c) R_{eq} , as $R = 20\Omega$ [1]

Determine

- a) R_{eq} , when $R = 0$ [1]
- b) R_{eq} when $R = \infty$ [1]
- c) R_{eq} , when $R = 20\Omega$ [1]

Vraag 1.3

Question 1.3

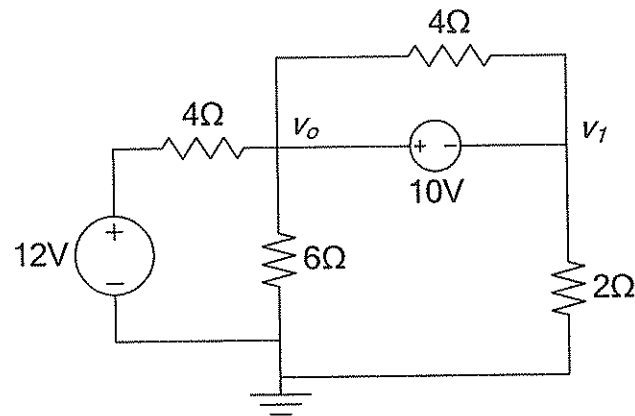


Skakel die stroombaan om van Δ na Y. Bereken R_1 , R_2 , R_3 , en R_{xy} . [4]

Convert the circuit from Δ to Y. Calculate R_1 , R_2 , R_3 , and R_{xy} . [4]

Vraag 1.4

Question 1.4



Gebruik NODALE analyse:

- a) Pas "KCL" en "KVL" toe om die volgende vergelykings te bepaal: [2]

$$a_1 v_o + a_2 v_1 = a_3 \quad (KCL)$$

$$b_1 v_o + b_2 v_1 = b_3 \quad (KVL)$$

(Bereken $a_1, a_2, a_3, b_1, b_2, b_3$)

- b) Bepaal v_o . [2]

Use NODAL analysis:

- a) Apply KCL and KVL to determine the equations in the form below: [2]

$$a_1 v_o + a_2 v_1 = a_3 \quad (KCL)$$

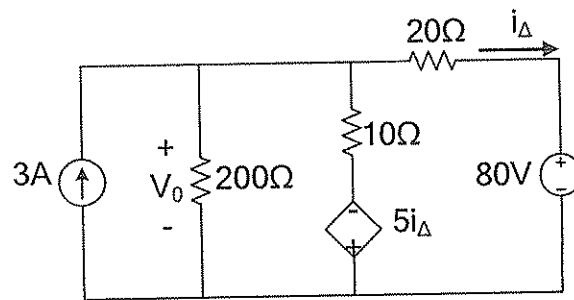
$$b_1 v_o + b_2 v_1 = b_3 \quad (KVL)$$

(Calculate $a_1, a_2, a_3, b_1, b_2, b_3$)

- b) Determine v_o . [2]

Vraag 1.5

Question 1.5

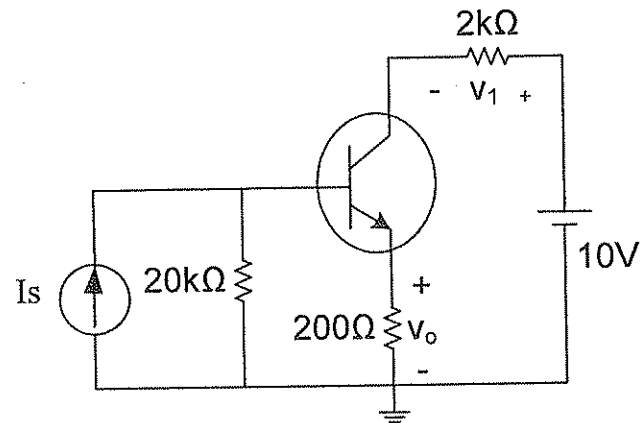


Gebruik node-analise om die stroombaan op te los. Wat is die waarde van v_0 ? [3]

Use nodal analysis to solve the circuit. What is the value of v_0 ? [3]

Vraag 1.6

Question 1.6



$\beta = 120$ en $v_{BE} = 0.75V$ vir die transistor in bostaande figuur.

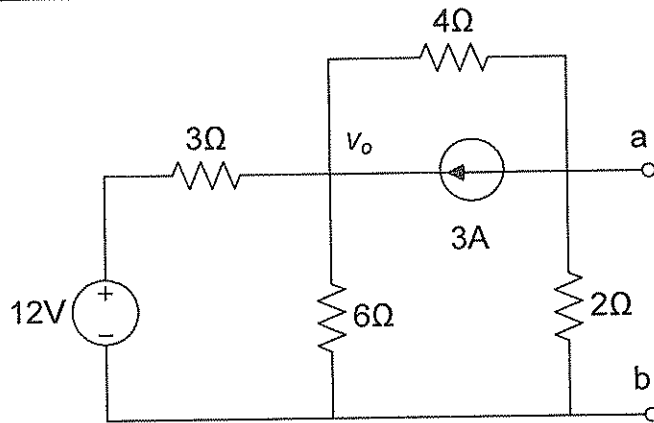
- a) Bepaal I_s as $v_o = 4.5V$, en skryf jou antwoord in μA . [2]
 b) Bepaal v_1 as $v_o = 3V$. [2]

For the transistor in the figure above, $\beta = 120$ and $v_{BE} = 0.75V$.

- a) Determine I_s if $v_o = 4.5V$, and express your answer in μA . [2]
 b) Determine v_1 if $v_o = 3V$. [2]

Vraag 1.7

Question 1.7

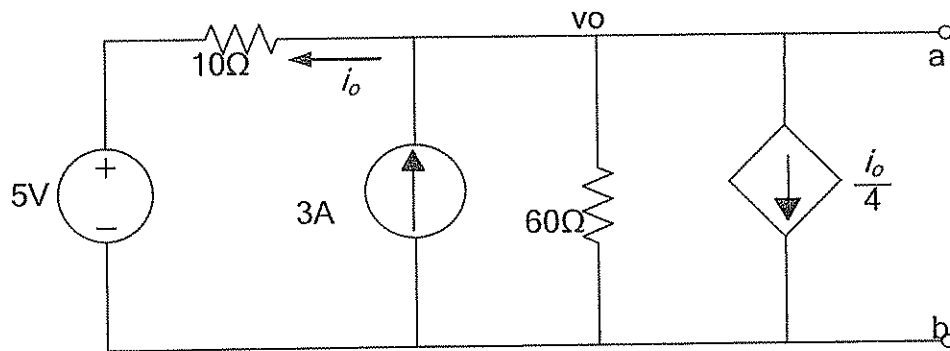


- (a) Bepaal R_{th} met verwysing tot terminale a–b. [1]
 (b) Bepaal die bydrae van die 12V spanningsbron tot die ope-baan spanning v_{ab} . [1]
 (c) Bepaal die bydrae van die 3A stroombron tot die ope-baan spanning v_{ab} . [2]

- (a) Find R_{th} with respect to terminals a–b. [1]
 (b) Find the contribution of the 12V voltage source to the open-circuit voltage v_{ab} . [1]
 (c) Find the contribution of the 3A current source to the open-circuit voltage v_{ab} . [2]

Vraag 1.8

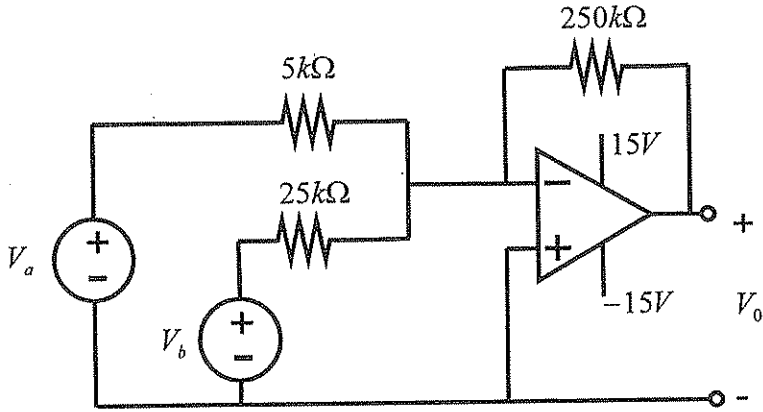
Question 1.8



Bepaal die Thevenin-ekwivalente weerstand R_{TH} met verwysing tot terminale a-b. [3]

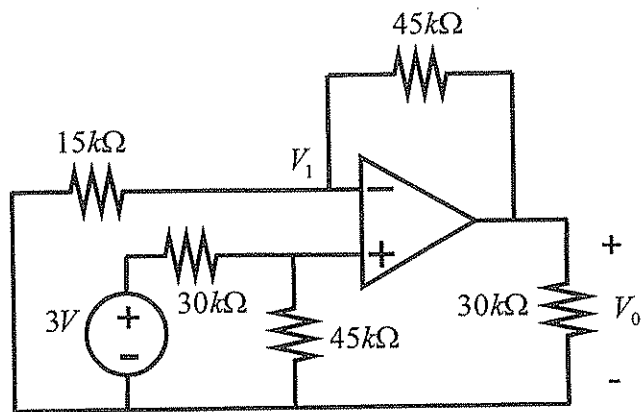
Determine the Thevenin equivalent resistance R_{TH} with respect to terminals a-b. [3]

VRAAG 2 – QUESTION 2 [12]

Vraag 2.1	Question 2.1
	
(a) Bepaal V_o as $V_a = 0.1 \text{ V}$ en $V_b = 0.25 \text{ V}$ [2] (b) As $V_b = 0.5 \text{ V}$, hoe groot kan V_a wees voordat die OV versadig? [2]	(a) Find V_o if $V_a = 0.1 \text{ V}$ and $V_b = 0.25 \text{ V}$ [2] (b) If $V_b = 0.5 \text{ V}$, how large can V_a be before the op-amp saturates? [2]

Vraag 2.2

Question 2.2



- (a) Bepaal V_1 [1]
 (b) Bepaal V_0 [2]

- (a) Determine V_1 [1]
 (b) Determine V_0 [2]

Vraag 2.3	Question 2.3
Die spanning oor die terminale van 'n $0.6 \mu\text{F}$ kapasitor is 0V vir $t < 0$ en $40e^{-15000t} \text{ V}$ vir $t \geq 0 \text{ s}$. Bepaal (a) $i(0)$ [2] (b) die energie gestoor in die kapasitor by $t = 0.1 \text{ ms}$. [1]	The voltage at the terminals of a $0.6 \mu\text{F}$ capacitor is 0V for $t < 0$ and $40e^{-15000t} \text{ V}$ for $t \geq 0 \text{ s}$. Determine (a) $i(0)$ [2] (b) the energy stored in the capacitor at $t = 0.1 \text{ ms}$. [1]

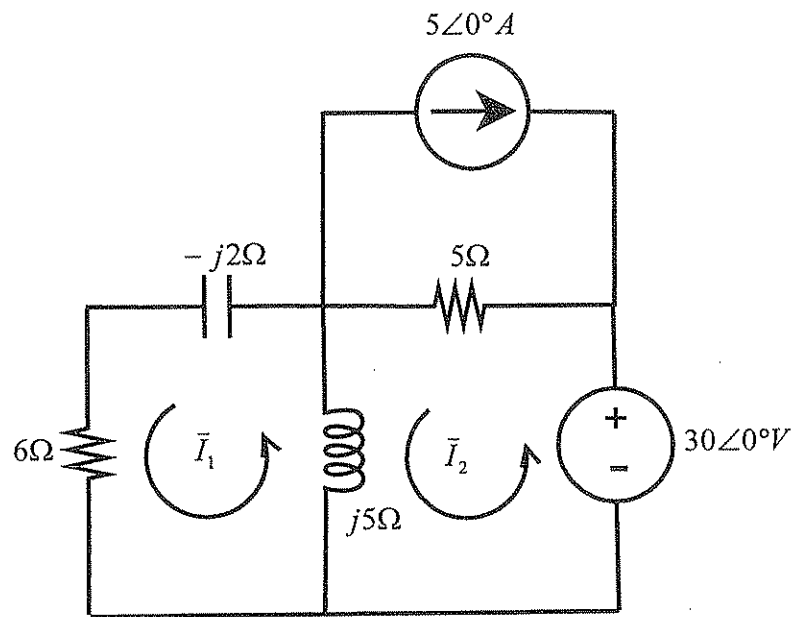
Vraag 2.4	Question 2.4
Die stroom deur 'n $100 \mu\text{H}$ induktor is $10 \sin(200t) \text{ A}$ vir $t \geq 0 \text{ s}$. Bepaal die drywing gelewer aan die induktor by $t = 10 \text{ ms}$. [2]	The current in a $100 \mu\text{H}$ inductor is $10 \sin(200t) \text{ A}$ vir $t \geq 0 \text{ s}$. Determine the power delivered to the inductor at $t = 10 \text{ ms}$. [2]

VRAAG 3 – QUESTION 3 [18]

Vraag 3.1	Question 3.1
(a) Skryf as 'n enkele cosinus uitdrukking $v(t) = 50\cos(\omega t + 30^\circ) + 30\cos(\omega t - 90^\circ)$ V [2]	(a) Write as a single cosine expression $v(t) = 50\cos(\omega t + 30^\circ) + 30\cos(\omega t - 90^\circ)$ V [2]
(b) Bepaal die ooreenstemmende fasor (in polêre vorm) $v(t) = 8\sin(10t + 70^\circ)$ V [1]	(b) Find the phasor (in polar form) corresponding to $v(t) = 8\sin(10t + 70^\circ)$ V [1]
(c) Bepaal die ooreenstemmende fasor (in polêre vorm) $i(t) = -60\cos(30t + 10^\circ)$ mA [1]	(c) Find the phasor (in polar form) corresponding to $i(t) = -60\cos(30t + 10^\circ)$ mA [1]
(d) Watter van die volgende fasors loop voor? $\bar{V}_1 = 10\angle -30^\circ$, $\bar{V}_2 = 15\angle 210^\circ$ [1]	(d) Which of the following phasors is leading? $\bar{V}_1 = 10\angle -30^\circ$, $\bar{V}_2 = 15\angle 210^\circ$ [1]
(e) Evalueer die determinant $\begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$ [2]	(e) Evaluate the determinant $\begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$ [2]

Vraag 3.2

Question 3.2



Gebruik lus-analise om twee gelyktydige vergelykings vir fasors $\bar{I}_1$ en $\bar{I}_2$ op te stel. Skryf in die vorm

$$\bar{a}\bar{I}_1 + \bar{b}\bar{I}_2 = 0$$

$$\bar{c}\bar{I}_1 + \bar{d}\bar{I}_2 = \bar{e}$$

met $\bar{a}$, $\bar{b}$, $\bar{c}$, $\bar{d}$ en $\bar{e}$ komplekse koeffisiënte in reghoekige vorm. [4]

Use mesh analysis to set up two simultaneous equations for phasors $\bar{I}_1$ and $\bar{I}_2$. Write in the form

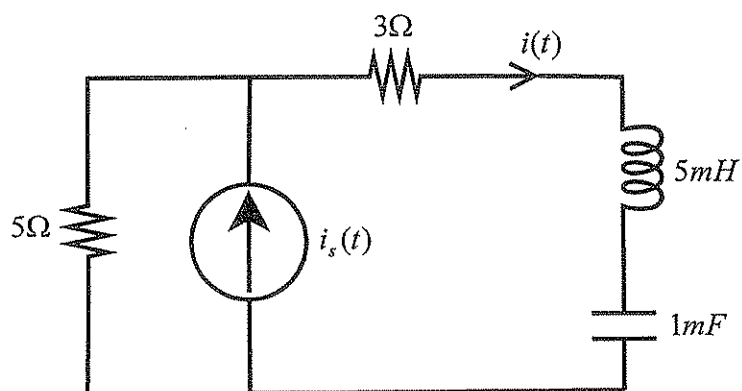
$$\bar{a}\bar{I}_1 + \bar{b}\bar{I}_2 = 0$$

$$\bar{c}\bar{I}_1 + \bar{d}\bar{I}_2 = \bar{e}$$

with $\bar{a}$, $\bar{b}$, $\bar{c}$, $\bar{d}$ and $\bar{e}$ complex coefficients in rectangular form. [4]

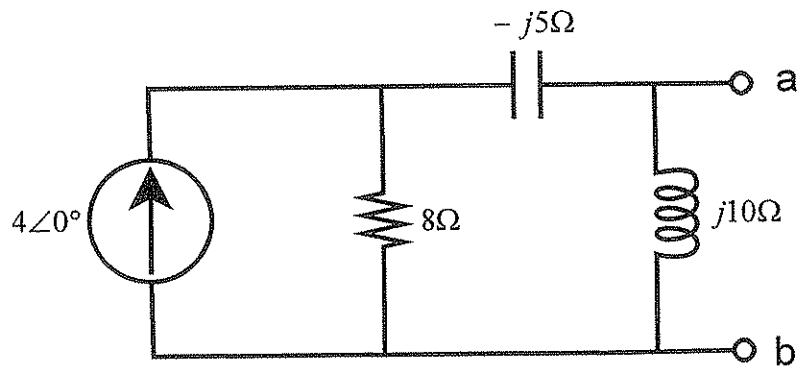
Vraag 3.3

Question 3.3



Gebruik brontransformaties om $i(t)$ te bepaal as $i_s(t) = 16\sin(200t + 30^\circ)$. [3]

Use source transformations to find $i(t)$ if $i_s(t) = 16\sin(200t + 30^\circ)$. [3]



Bepaal die Norton ekwivalent tov terminale a-b. Skryf die antwoorde in polêre fasorvorm. [4]

Find the Norton equivalent wrt terminals a-b. Write the answers in polar phasor form. [4]

EINDE - END